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Microsoft Power BI

1 Data Loading

Data Loading is the first and one of the most important stages of any Microsoft Power BI project. Before performing data cleaning, transformation, modelling, DAX calculations, or dashboards development, the data must first be imported correctly into Power BI Desktop.

Power BI supports importing data from hundreds of different sources including Microsoft Excel, CSV files, Text files, SQL Server, Oracle Database, MYSQL, PostgreSQL, SharePoint, Folder, Web APIs ,Azure Services, SAP, Salesforce, Google Analytics, and many cloud-based services.

Practice Questions

1.1 ABC Retail Pvt. Ltd. has provided an Excel workbook named Sales.xlsx. Import this workbook into Microsoft Power BI Desktop.

Solution:

- Open Microsoft Power BI Desktop.
- Click Get Data.
- Select Excel Workbook.
- Browse to Sales.xlsx.
- Click Open.
- Select the required worksheet.
- Click Load.

1.2 Import a CSV file named Products.csv into Power BI Desktop.

Solution:

- Click Get Data.
- Choose Text/CSV.
- Select Products.csv.
- Click Open.
- Verify delimiter.
- Click Load.

1.3 Preview data before importing it into Power BI.

Solution:

- Open Get Data.
- Select any data sources.
- Choose the dataset.
- Observe the Preview window.
- Verify records.
- Click Load.

1.4 Load data from a folder containing twelve monthly Sales CSV file.

Solution:

- Click Get Data.
- Select Folder.
- Browser to the Sales folder.
- Click OK.
- Click Combine and Load.

1.5 Connect Power BI Desktop to a Microsoft SQL Server database named RetailDB.

Solution:

- Click Get Data.
- Select SQL Server.
- Enter the Server Name.
- Enter the Database Name.
- Select Import Mode.
- Authenticate.
- Select required tables.
- Click Load.

1.6 Verify that all imported columns have the correct data types before beginning report development.

Solution:

- Open Data View.
- Select each table.
- Review every column.
- Verify text column.
- Verify Whole Number columns.
- Verify Decimal Number columns.
- Verify Date columns.
- Correct any incorrect data types.

1.7 Import data directly from a web page containing a product price table.

Solution:

- Click Get Data.
- Select Web.
- Enter the webpage URL.
- Click OK.
- Select the required HTML.
- Click Load.

1.8 Verify that all imported tables are successfully loaded before beginning data transformation.

Solution:

- Open the Fields pane.
- Confirm all required tables are listed.
- Open Data View.
- Verify that each table contains records.
- Check row counts and column names.

1.9 Connect Power BI Desktop to a SharePoint Folder containing monthly sales reports.

Solution:

- Click Get Data.
- Select SharePoint Folder.
- Enter the SharePoint site URL.
- Authenticate using organizational credentials.
- Select the required files.
- Click Load.

1.10 Verify that all imported datasets are ready for the next stage of Power BI development.

Solution:

- All required tables are imported.
- Column names are connected.
- Data types are correct.
- No duplicate tables exist.
- Row counts appear reasonable.
- No empty datasets.
- File connections are working.
- Required columns are available.
- Tables are properly renamed.
- Dataset is ready for Power Query.

1.11 Import the Sales.xlsx workbook, but instead of loading it directly, open it in Power Query Editor to review and prepare the data before loading.

Solution:

- Open Power BI Desktop.
- Click Get Data.
- Select Excel Workbook.
- Choose Sales.xlsx
- In the Navigator window, select the Sales table.
- Click Transform Data.
- Review the dataset in Power Query Editor.
- Click Close & Apply.

1.12 Import data from a folder containing monthly Excel sales reports and automatically combine them into a single table.

Solution:

- Click Get Data.
- Select Folder.
- Browse to the Sales Reports folder.
- Click OK.
- Choose Combine & Transform Data.
- Verify the sample file.
- Apply the changes.
- Click Close & Apply.

1.13 Change the file path of an imported Excel workbook after it has been moved to another folder.

Solution:

- Open Power Query Editor.
- Select the required query.
- Click the Source step.
- Browse to the new file location.
- Save the changes.
- Click Close & Apply.
- Refresh the dataset.

1.14 Import data from Microsoft Excel while selecting only specific columns required for reporting.

Solution:

- Import the Excel workbook.
- Open Power Query Editor.
- Select the query.
- Click Choose Columns.
- Select only the required columns.
- Remove the remaining columns.
- Click Close&Apply.

1.15 Import multiple CSV files that use different delimiters and ensure the correct delimiter is selected during import.

Solution:

- Open the CSV file using Get Data.
- Review the preview window.
- Verify the detected delimiter.
- Change the delimiter manually if required.
- Confirm that all columns are correctly separated.
- Click Load.

1.16 Connect to an OData Feed and import customer information into Power BI.

Solution:

- Click Get Data.
- Select OData Feed.
- Enter the service URL.
- Authenticate if required.
- Select the Customer table.
- Click Load.

1.17 Use Query Dependencies in Power Query to understand how imported queries are connected.

Solution:

- Open Power Query Editor.
- Navigate to the View tab.
- Click Query Dependencies.
- Study the dependency diagram.
- Identify source queries and dependent queries.

1.18 Prepare an imported dataset for the next phase of the Power BI development lifecycle.

Solution:

- Verify all required tables are imported.
- Check table names.
- Validate column names.
- Verify data types.
- Remove unnecessary tables.
- Confirm successful refresh.
- Check for duplicate datasets.
- Review Applied Steps.
- Verify source connection.
- Save the Power BI report.

1.19 Import a folder containing monthly Sales Excel workbooks where new files are automatically added every month. Configure the query so that future files are included automatically after refresh.

Solution:

- Select Get Data.
- Choose Folder.
- Browse to the Sales folder.
- Select combine & transform.
- Verify the sample file.
- Apply the transformation.
- Save the report.
- Add a new monthly workbook to the folder.
- Click Refresh.

1.20 A report connects to three different data sources: Excel, SQL Server, and SharePoint. Configure the report so that all sources refresh successfully.

Solution:

- Connect each source individually.
- Configure authentication.
- Verify credentials.
- Import required tables.
- Save the report.
- Refresh all datasets.
- Resolve any authentication errors.

1.21 Your organization stores daily sales files in a folder. Occasionally, users accidentally save files with missing columns or different column names. Design a robust data loading strategy to handle this situation.

Solution:

- Import the folder using the Folder connector.
- Use Combine & Transform.
- Compare the sample file with incoming files.
- Validate column names.
- Handle missing columns using Power Query transformations.
- Remove invalid files if required.
- Apply the changes.

1.22 Design a production-ready enterprise data loading framework for a Power BI reporting solution.

Solution:

- Identify business requirements.
- Verify data sources.
- Authenticate securely.
- Import only required datasets.
- Review Applied Steps.
- Validate data quality.
- Verify refresh.
- Document source information.
- Save the report.
- Hand over for the Data Modeling.

1.23 After publishing a Power BI report, scheduled refresh begins to fail due to authentication issues. Develop a troubleshooting process.

Solution:

- Review refresh history.
- Identify the failing data sources.
- Verify credentials.
- Re-enter authentication details if required.
- Confirm source availability.
- Run a manual refresh.
- Monitor the next scheduled refresh.

1.24 Perform a complete technical review of a Power BI project's data loading process before handing it over to the Data Modeling team.

Solution:

- Verify all source connections.
- Validate credentials.
- Review Applied Steps.
- Verify data types.
- Confirm row counts.
- Review Query Dependencies.
- Remove unused queries.
- Verify naming conventions.
- Test refresh.
- Save documentation.

1.25 Design a complete enterprise-ready Data Loading Framework that can serve as the standard for all future Power BI projects.

Solution:

- Gathering business requirements.
- Identify all data sources.
- Select the appropriate connection method.
- Configure authentication securely.
- Import only required tables and columns.
- Create staging queries.
- Perform data validation.
- Verify data types.
- Review Applied steps.
- Validate Query Dependencies.
- Disable loading for intermediate queries.
- Test dataset refresh.
- Document the complete ETL process.
- Obtain peer review.
- Approve the dataset for Data Modeling.